



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

Consult 2: CA6121 AI in Project Management

Group Project Steps & Progress

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Recap



AI in Project Management or AI Project?

High level Perspective, introduction



Artificial Intelligence and Project Management

Recap

Our focus in this course



AI in Project
Management



AI Project



Pexels.com



Project Integration (recap)

AI in Project Management



Project Execution Strategies: Project Management Processes

Project management processes:

1.Initiating

2.Planning

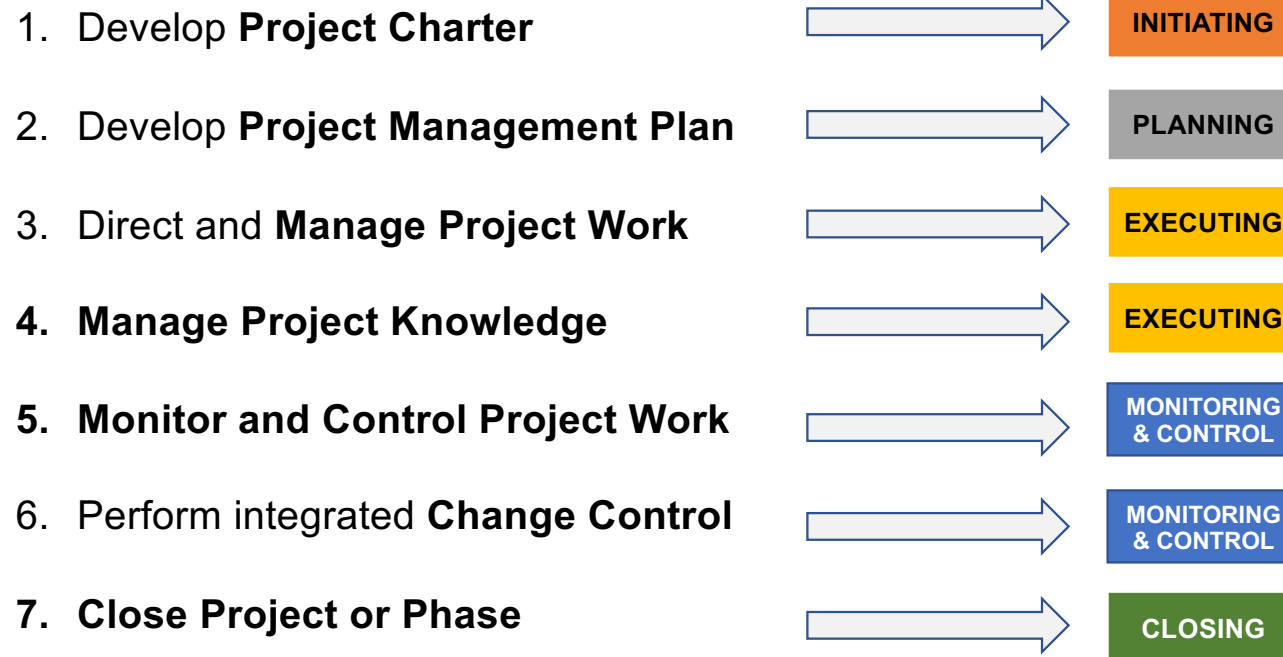
3.Executing

4.Monitoring and Controlling

5.Closing



Project Integration Management - Processes





Project Execution Strategies: Project Management Knowledge

Project management knowledge draws following areas:

- | | |
|---------------|---------------------------|
| 1.Integration | 6. Procurement |
| 2.Scope | 7. Human resources |
| 3.Time | 8. Communications |
| 4.Cost | 9. Risk management |
| 5.Quality | 10.Stakeholder management |



What it takes to use AI in Projects?

AI in Project Management



AI in Project Management

- Understanding of Project Management Workflow and where it adds value to use AI?
- AI can not be implemented on every workflow.
- Some of the areas can be used as:
 - Data consolidation, analysis, collation
 - Generating periodic communications
 - Auto response to regular queries
 - Planning
 - Reporting
 - Monitoring
 - Prediction
 - Others...



AI in Project Management: What is takes to implement AI?

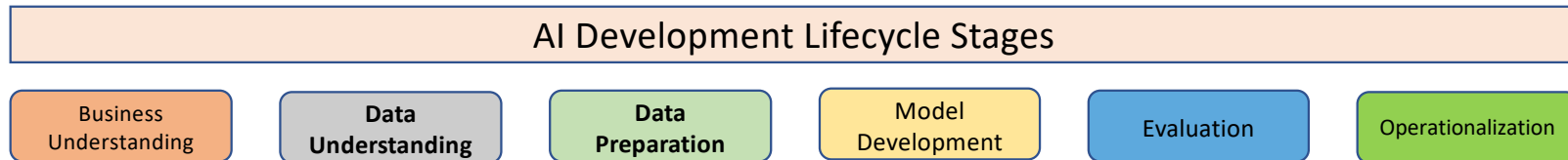
Six (6) phases of implementation process of a AI project:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Model Development
5. Evaluation
6. Operationalization



AI Development Projects

Recap



Project Manager could be from either of the disciplines:

- Computer Science
- Data Science
- Information Technology
- Other discipline with knowledge and interest in data



AI in Project Management: Pitfalls

Data quality and availability

Finding relevant, up-to-date, and clean data is a challenge.

- Data redundancy and gaps
- Unsupported formats
- Unclear permission and ownership issues of data



AI in Project Management: Managing Data

Data quality and availability

- Verify data sources
- Mindful of data limitations and biases
- Set-up alerts for anomalies
- Monitor and validate



AI in Project Management: AI Models

Choosing and Integrating AI Models

Picking the right AI model and integrating the existing workflows can be challenging.

- Different workflows
- Deployment challenges



AI in Project Management: AI Models

Choosing and Integrating AI Models

- Assess project needs
- Set clear expectations
- Pilot test
- Collaborate
- Road map



AI Tools in Project Management

High level Perspective



PM knowledge areas and suitable types of AI tools.

From lecture

Item	PM knowledge area	Dominant knowledge type	Most suitable types of AI tools	Examples of AI capabilities
1	Integration management	- Formal - Data Driven - Tacit	GenAI AI Agents (LLMs & Integration)	Decision-making support based on real-time tracking and deviation alerts.
2	Scope management	- Data Driven - Tacit	GenAI AI Agents (LLMs & Integration)	Drafting work breakdown structures, clarifying deliverables
3	Schedule management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Timeline generation, scheduling, forecasting, Critical Path identification.
4	Cost management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Budget forecasting, variance analysis, EVM
5	Quality management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Quality monitoring, anomaly detection.
6	Resource management	- Formal - Data Driven	ML for optimization AI Agents (LLMs & Integration)	Resource allocation, workload balancing.
7	Communication management	- Formal - Tacit	GenAI AI Agents (LLMs & Integration)	Summarizing meetings, drafting reports, language translation.
8	Risk management	- Data Driven - Tacit	GenAI ML for optimization AI Agents (LLMs & Integration)	Identifying emerging risks, probabilistic impact analysis.
9	Procurement management	Formal	Automation AI Agents (LLMs & Integration)	Contract analysis, vendor, sub-con selections.
10	Stakeholder management	- Formal - Tacit	GenAI AI Agents (LLMs & Integration)	Drafting engagement plans, analyzing feedback, conflict resolution assistance.



Student Exercise 1- Self study no submission required

From lecture

- Perform a research to find and list the available AI tools which may be used in the project management processes.
 - Collating the business documentation.
 - Reviewing the lessons from the past projects.
 - Data quality check and filtering.
 - Planning and scheduling activities.
 - Resource Management.
 - Risk Management.
 - Cost and EVM (Earned Value Management).
 - Project Reporting.
 - Project Communication.
 - Any other tool which you may feel could be used.



Project Topic and Approach

AI in Project Management



AI in Project Management: Select Project Title

Recap

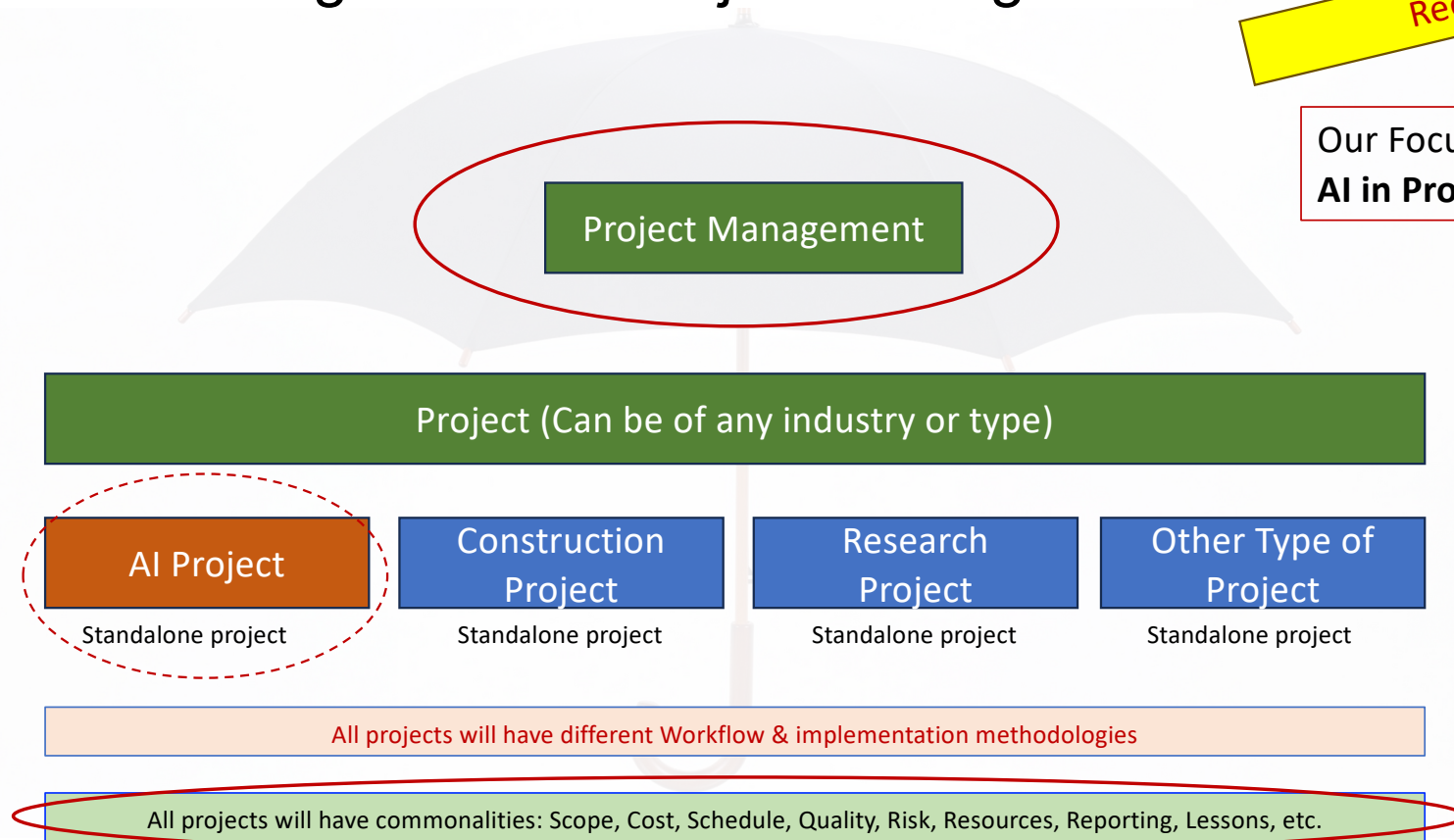
- Your team needs to select a project title.
- Project can be from any of the following areas:
 - IT / Software development project
 - Construction or building/ manufacturing plant project (new or modification).
 - Research project.
 - Any other project of your choice.
- Project title must be submitted for in-principal approval.



Artificial Intelligence and Project Management

Recap

Our Focus
AI in Project Management





AI in Project Management:

- Criteria to select a project for group assignment:
 - Can have either or all phases of project covered in the lecture
 - Initiation,
 - planning,
 - executing,
 - monitoring & control,
 - close-out.
- Must draw a workflow of the current situation and proposed improvement.
- **Improve process** for a project by implementing AI tool of your choice. (Example: collating of notes of meeting, preparation of EVM, sorting the team profiles, etc.).



AI in Project Management: Where can it help a Project Manager?

INITIATING

- Project Business Case development.
- Review and analyze the contract document.
- Client supplied technical and commercial documents compile and analysis.
- Project Charter development.

PLANNING

- Project Management Plan preparation and updating.
- Project specific procedures.
- Project compliance to contract, local regulations, etc.

EXECUTING

- Team selection (Review of applications).
- Project schedule development and updates.
- Earned Value Management (EVM) and Forecasting.
- Budget vs Actual performance analysis.
- Notes of meeting automation, summary, issue to the stakeholders.
- Issue regular communication with the stakeholders. (Emails, Project Updates, Newsletters, etc.).
- Periodic project reports (weekly / monthly / quarterly).
- Project Quality and Inspection (Plan and Actual performance / analysis).
- Project Risk Analysis (Plan and Actual performance / analysis).
- Project Change Management and its impact on project. (Plan and Actual performance / analysis).
- Project procurement management (vendor, sub-con selection, tender comparison, recommendation, performance analysis, etc.).
- Expenses reporting.

MONITORING & CONTROL

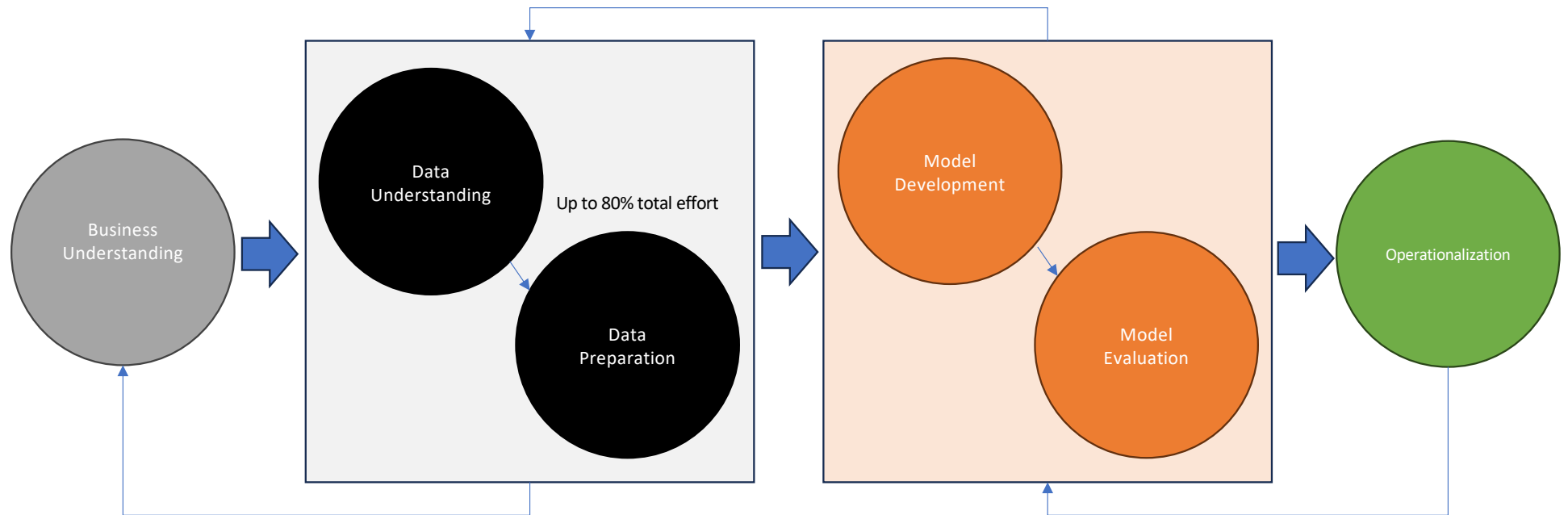
CLOSING

- Planned vs Actual Data analysis (Scope, Schedule, Cost, Resources).
- Changes and rework analysis.
- Quality/ Safety/ Compliance/ Testing/ Deployment/ Handing over, performance and analysis.
- Vendors / Sub-contractors/ Suppliers performance analysis.
- Team performance analysis.
- Lessons mapping and analysis.
- Project close-out documentation.

All of the above mentioned can be improved using AI Tools



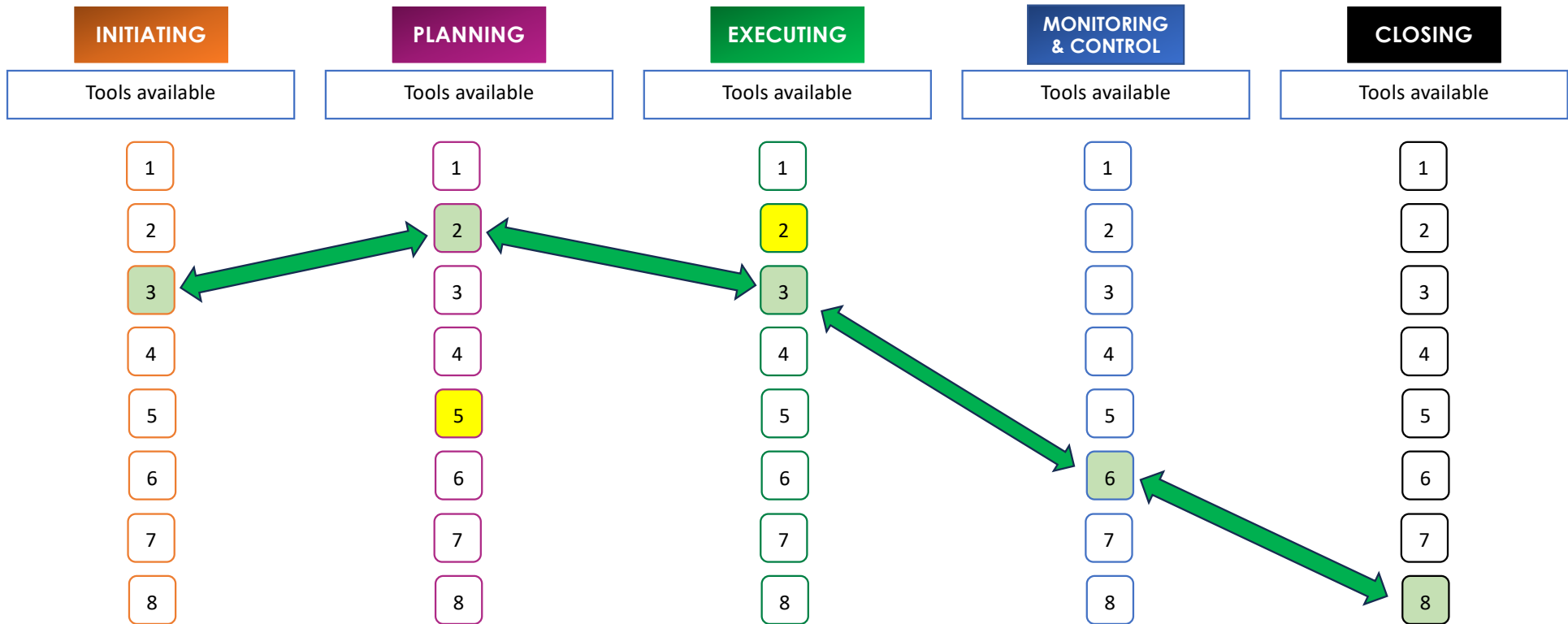
AI tools in Project Management: General Workflow





Tutorial 2 will have hands-on exercise on this

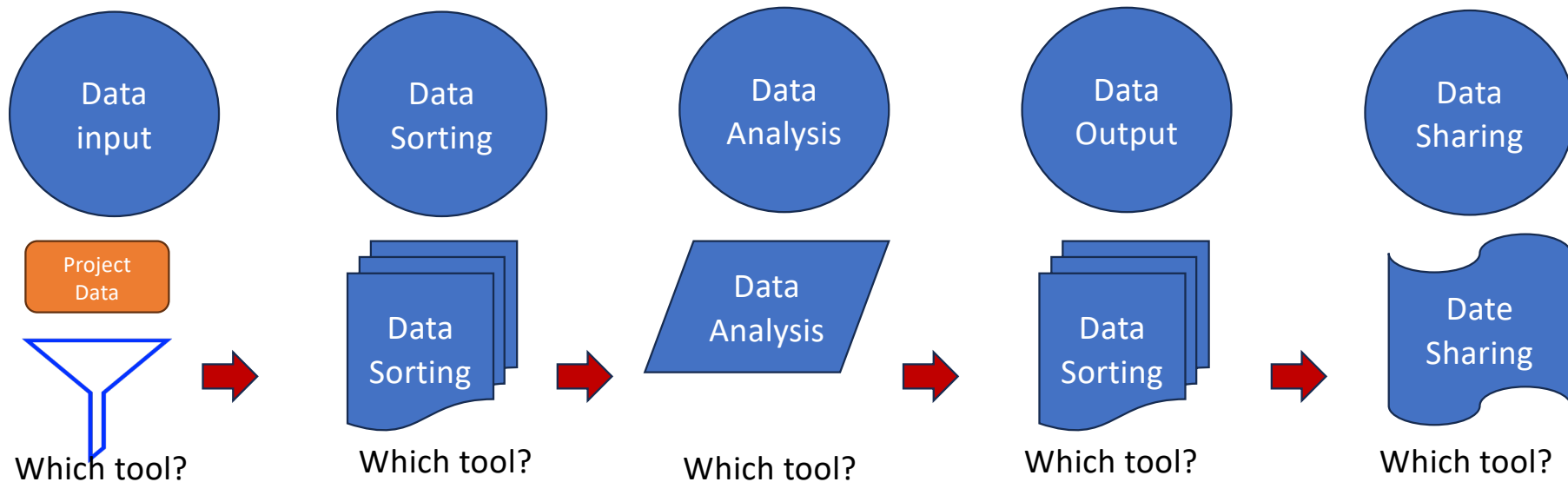
AI in Project Management: Select the right AI Tools





Tutorial 2 will have hands-on exercise on this

AI in Project Management: **Workflow**





AI in Project Management:

- Use the consultation session regarding the group project.
- AI Tools can be of your choice.
- Marking will have the rubrics to include:
 - Group report (content and quality).
 - Workflow snapshot.
 - Proposed AI tools for the solution to the workflow.



AI in Project Management:

- Use the project report template provided.
- Please do not delete the headings and bullets. (please mark N.A, if any of them is not applicable for your selected project).
- Use the guidelines provided for preparation.



Group Project: What Next?

AI in Project Management



Topic understanding and to set a path to work on.

Client wish to implement AI Tools
in Project Management

Client

- You have project topic selected.
- Your inputs:
 - Workflow with **project steps** explained.
 - **Identify the areas** where AI Tools can be implemented in Project Management Processes.
 - **Resources needed** to develop and implement the AI solutions.
 - **Proposed Cost / Time / Schedule** to execute the project.
 - Any other **value-added** pointers.

Your Group Represents as Consultant
and expected to advise the client.

Consultant





Your Group Project Progress?

AI in Project Management



Your Group Project Progress?

1. Teamwork is important.
2. Appoint a group lead who will act as project manager to coordinate this group assignment activities.
3. Group assignment topic is reviewed and agreed by all team members?
4. Problem Statement development status?
5. Have you finalized the scope?
6. Where are you on the workflow development for the overall project?



Topic understanding and to set a path to work on.

- Understand your group project topic.
(Use tutorial 01 class exercise as reference).
- What **type of project and which industry workflow** your project belongs to?
(Suggestion: IT, Construction, Building, Manufacturing, Aerospace, etc.?)
- Explain **current problem in this workflow** where your group is trying to solve by implementing AI tools in Project Management Process?
(Example: Notes of meeting compilation, review, analysis, draft email to be sent to the attendees).
- What is your **proposal to solve** the identified problem statement?
(Example: AI tool implementation to collate meeting transcript and compile into notes of meeting, perform review, analysis, and prepare draft email to be sent to the attendees)
- Draw a **workflow or block diagram or write down the steps** you will take to solve.
- **Once topic is finalized**, ask again: is your team confident to work on this project topic or wish to revise or fine-tune?



What to do before next **consultation?**



Group topic submission

- Stay in touch with your group mates.
- Get to know each other.
- Explore the topics in depth for the group assignment do you wish to work on.
- Come up with **proposed topic** for your group.
- **Submit your group topic** one day before the next consultation class.

<https://forms.cloud.microsoft/r/B8zBtFa6Nb>



[CA6121 NTU Group Assignment Topic selection Mar2026 – Fill out form](https://forms.cloud.microsoft/r/B8zBtFa6Nb)



What to do before next tutorial class?



Tutorial Classes

- Ensure you have **watched the relevant lecture(s)**.
- **Complete the learning exercise, if any** suggested in the lecture.
- Please **bring your laptop** to participate in class exercises/ activities.



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End of class



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Batam, Indonesia

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